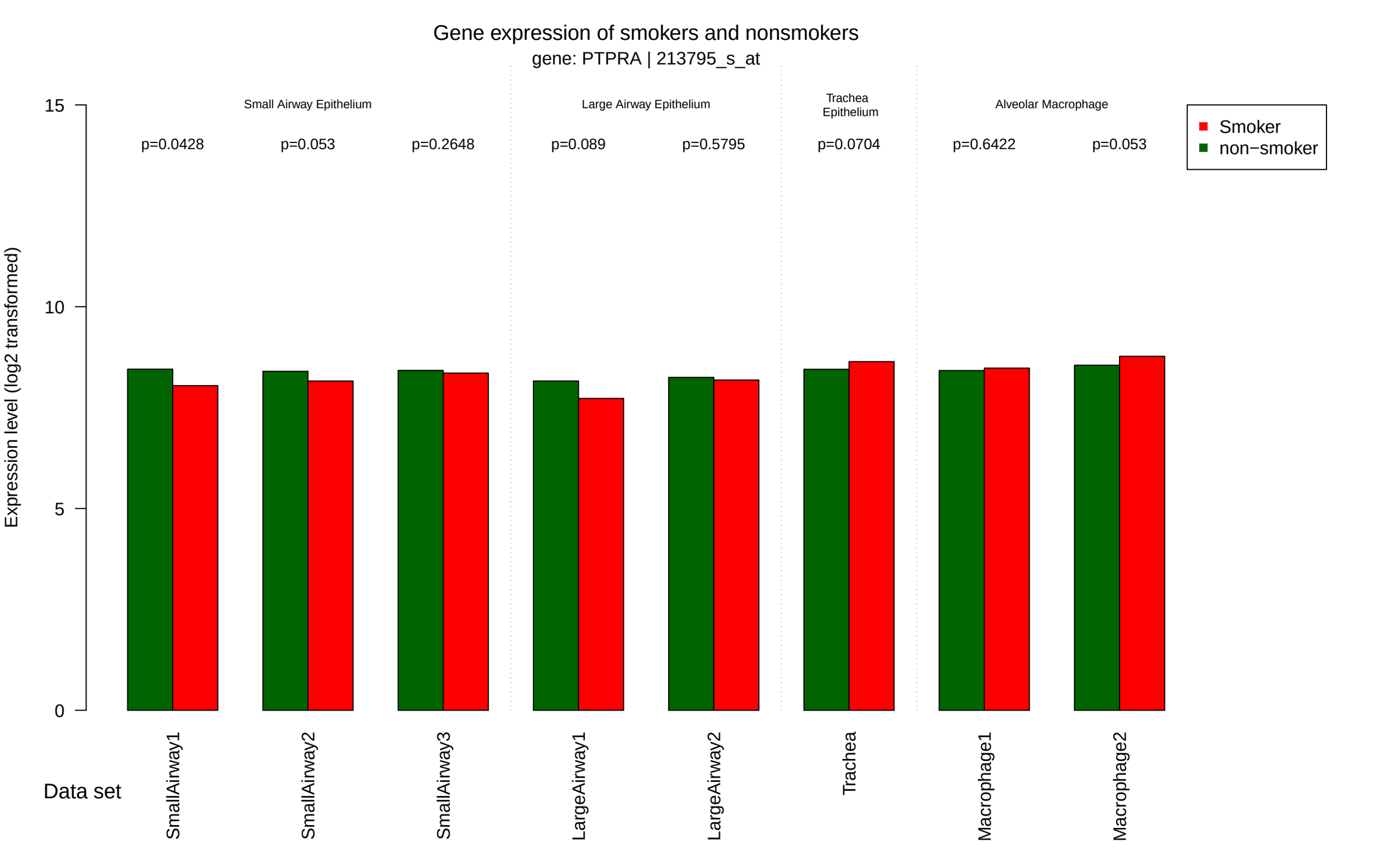
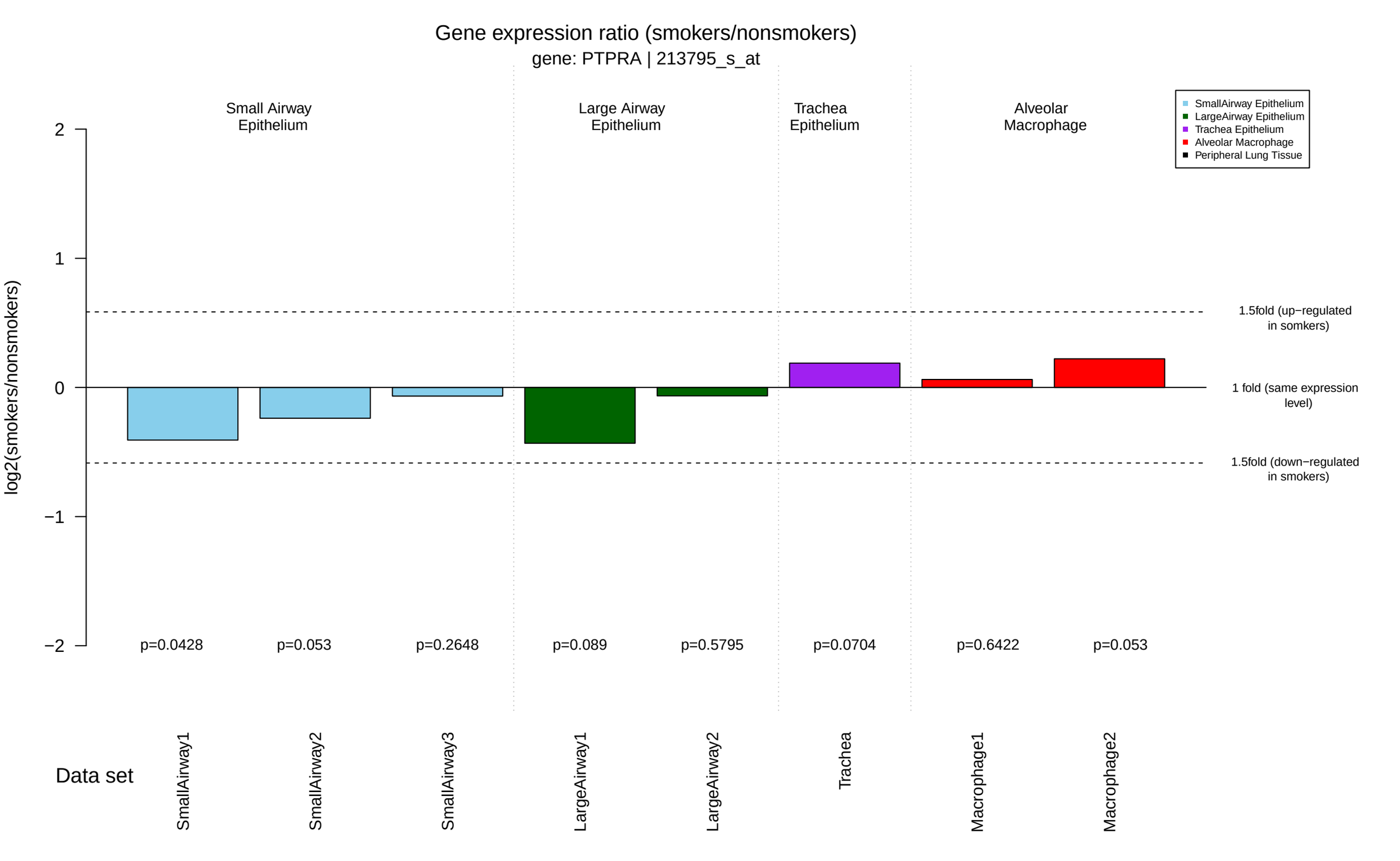
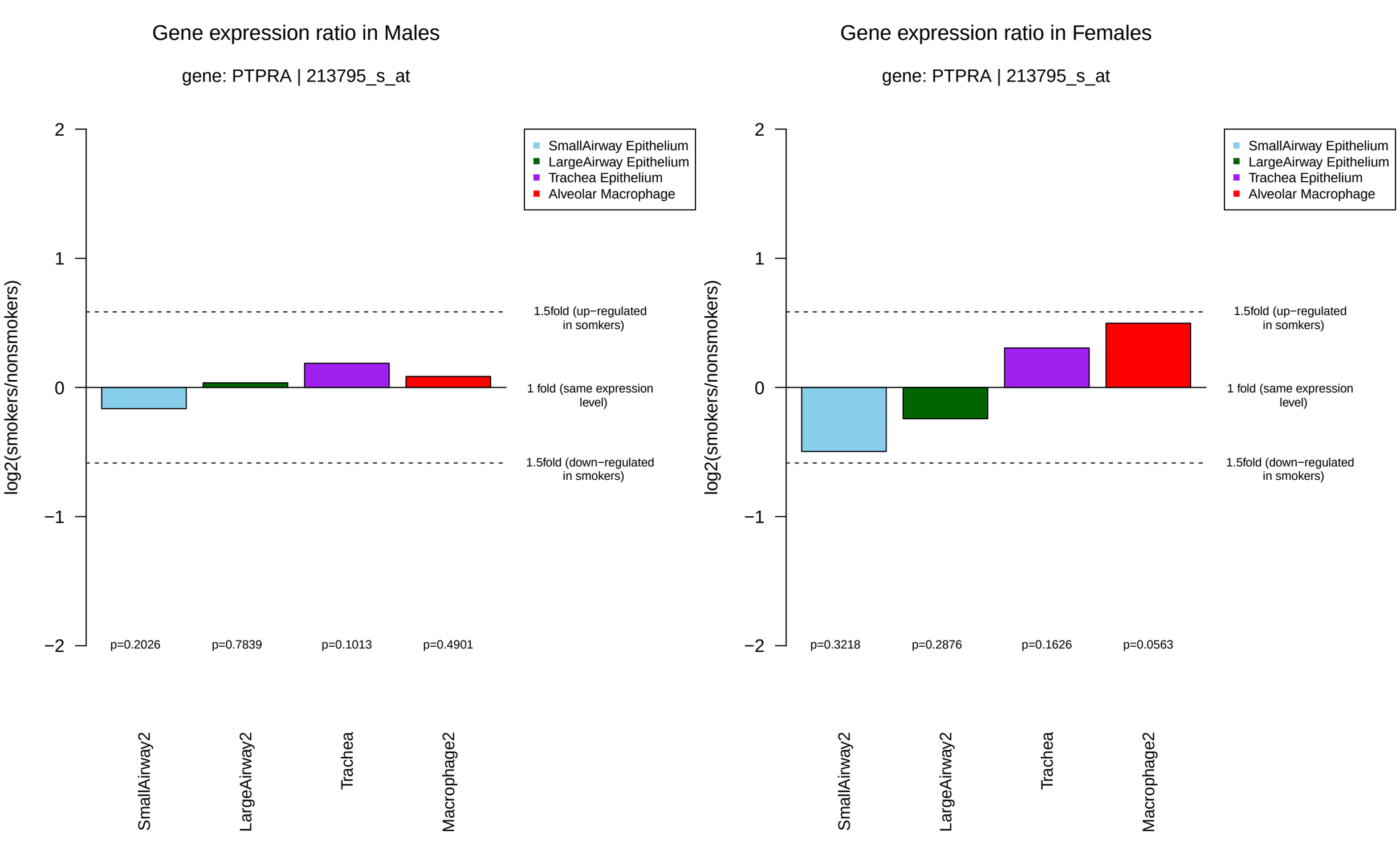
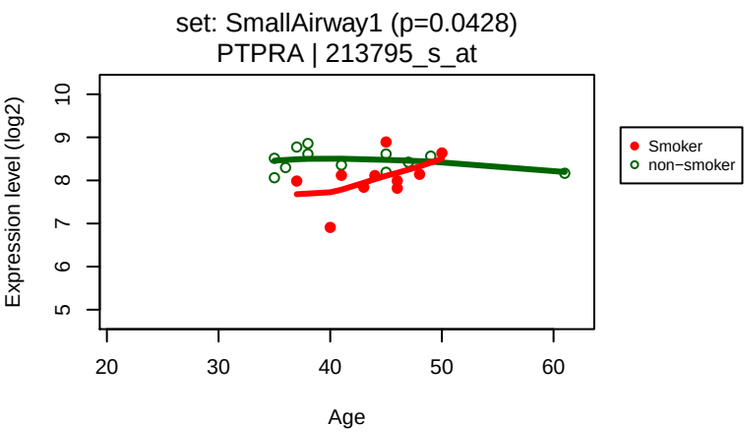
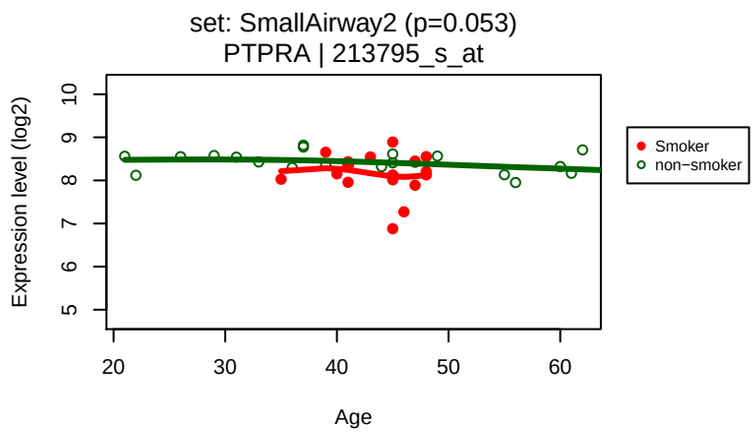
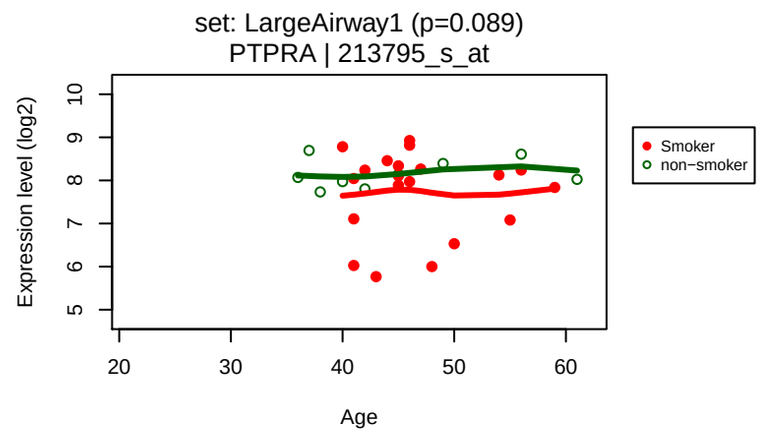

Gene expression of smokers and nonsmokers
gene: PTPRA | 213795_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0428
p=0.053
p=0.2648
p=0.089
p=0.5795
p=0.0704
p=0.6422
p=0.053
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

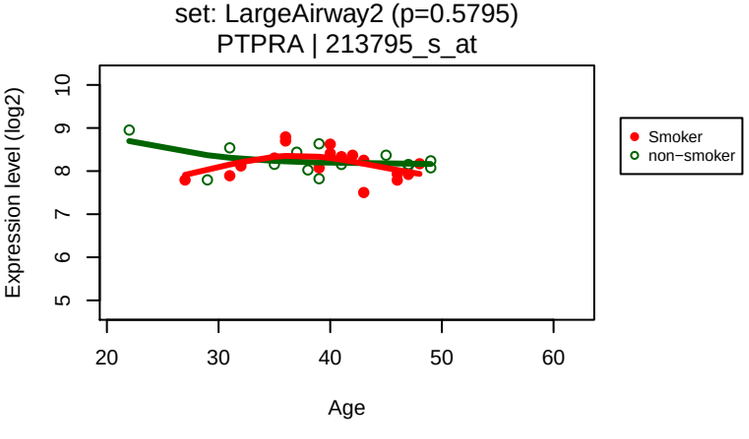

Gene expression ratio (smokers/nonsmokers)
gene: PTPRA | 213795_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0428
p=0.053
p=0.2648
p=0.089
p=0.5795
p=0.0704
p=0.6422
p=0.053
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

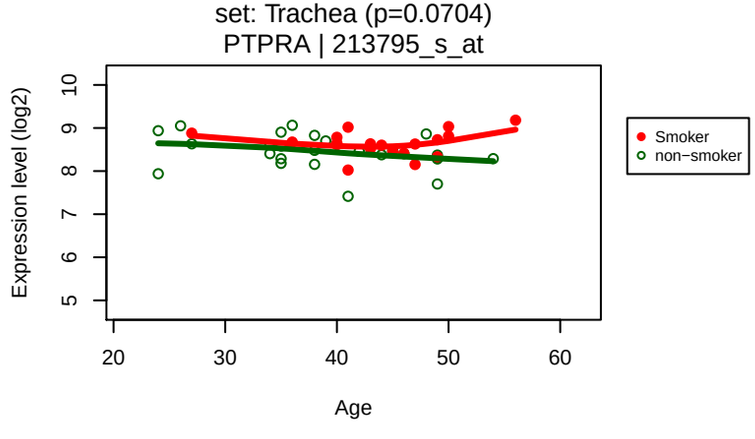

Gene expression ratio in Males
gene: PTPRA | 213795_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2026
p=0.7839
p=0.1013
p=0.4901
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPRA | 213795_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3218
p=0.2876
p=0.1626
p=0.0563
SmallAirway2
LargeAirway2
Trachea
Macrophage2

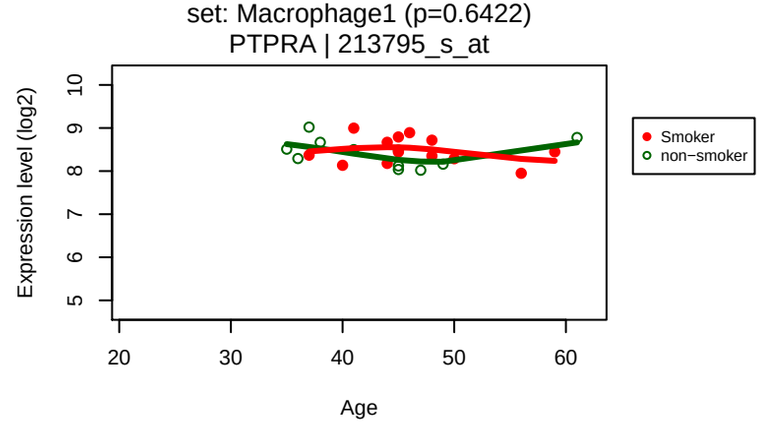

set: SmallAirway1 (p=0.0428)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker

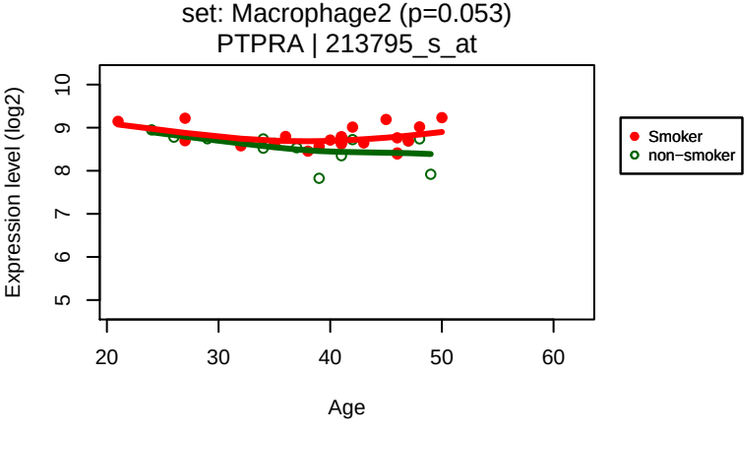

set: SmallAirway2 (p=0.053)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker

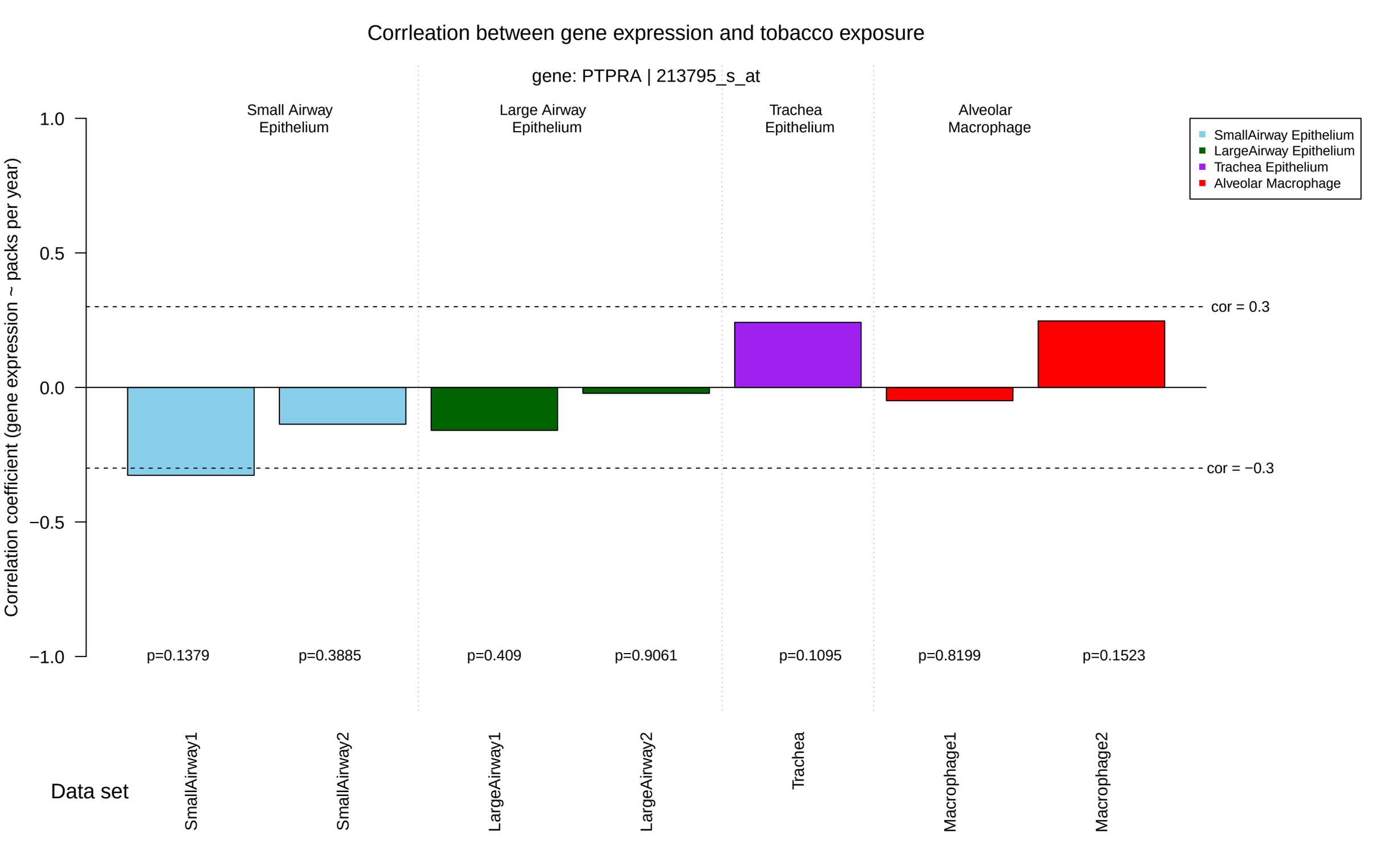

set: LargeAirway1 (p=0.089)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5795)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0704)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6422)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.053)
PTPRA | 213795_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPRA | 213795_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1379
p=0.3885
p=0.409
p=0.9061
p=0.1095
p=0.8199
p=0.1523
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2